

3-Mark Questions & Answers — MPCA Unit 3: The Intel Microprocessor 8086 Architecture

Q1. Draw and explain the functional pin diagram of 8086 (any 6-8 important pins).

1. **AD0 - AD15 (Pins 2-16, 39):** Multiplexed Address/Data lines carrying 16-bit data or lower 16-bit memory address.
2. **A16/S3 - A19/S6 (Pins 35-38):** Multiplexed upper 4-bit address lines and status signals.
3. **ALE (Address Latch Enable):** High pulse indicating valid address is present on multiplexed bus.
4. **RD / WR:** Memory/IO Read and Write control output signals.
5. **MN/MX' (Pin 33):** Configures CPU for Minimum Mode (+5V) or Maximum Mode (GND).
6. **RESET (Pin 21):** Resets CPU; sets **CS = FFFFH** , **IP = 0000H** .

Q2. Explain memory segmentation in 8086 with a diagram.

```

1 MB Memory Space (00000H to FFFFFH)
+-----+ FFFFFH
| UNMAPPED MEMORY SPACE |
+-----+
| EXTRA SEGMENT (ES)      (64 KB) | <- ES:DI
+-----+
| STACK SEGMENT (SS)      (64 KB) | <- SS:SP/BP
+-----+
| DATA SEGMENT (DS)       (64 KB) | <- DS:BX/SI
+-----+
| CODE SEGMENT (CS)        (64 KB) | <- CS:IP
+-----+ 00000H
  
```

- Total Physical Address Space = $2^{20} = 1 \text{ MB}$.
- Divided into 4 active 64KB logical segments: **CS** (Code), **DS** (Data), **SS** (Stack), **ES** (Extra).

Q3. Explain the concept of banking (memory banking) in 8086.

```

1MB PHYSICAL MEMORY
/
512 KB (D0 - D7)      512 KB (D8 - D15)
Connected to A0 = 0    Connected to BHE' = 0
  
```

- Memory is split into two 512KB banks:
- **Even Bank (Lower Bank):** Contains even byte addresses (**00000H, 00002H, ...**), driven by **A0 = 0** .
- **Odd Bank (Upper Bank):** Contains odd byte addresses (**00001H, 00003H, ...**), driven by **BHE' = 0** .
- Enables fetching 16-bit word data in a single memory cycle when aligned on an even address.

Q4. Differentiate between minimum mode and maximum mode operation of 8086.

Feature	Minimum Mode	Maximum Mode
Pin Setup	Pin 33 (MN/MX') connected to +5V.	Pin 33 (MN/MX') connected to GND.
System Architecture	Single CPU processor system.	Multi-processor / Co-processor system (8087 FPU, 8089 IOP).
Control Generator	8086 directly outputs RD , WR , ALE , M/IO .	External 8288 Bus Controller decodes status pins S0' , S1' , S2' .

Q5. Explain the need for demultiplexing the address/data bus in 8086.

1. Pins **AD0 - AD15** carry memory addresses during the first clock cycle (**T1**) and data during **T2, T3, T4** .
2. Without demultiplexing, memory chips would lose the target address when data values are placed on the pins during **T2** .
3. Latches (74LS373) latch the address when **ALE = 1** , preserving a stable 20-bit address throughout the memory cycle.